

1 Menger's Theorem

Theorem 1.1 (Menger's Theorem). *Let G be a finite undirected graph and A and B two disjoint subsets of vertices. Then the minimum number of vertices that need to be removed to disconnect A from B is equal to the maximum number of vertex-disjoint paths from A to B .*

Proof. On a graph G , define $n(G)$ as the number of vertex-disjoint paths between A and B , and the minimum cut as $m(G)$.

$m \geq n$: If we have $N = n(G)$ vertex-disjoint paths and we remove $M < n(G)$ of these vertices, A and B must still be connected through at least one path since no paths can share a vertex.

$n \geq m$: For any graph $G = (V, E)$, we prove by induction on induced subgraphs of size k (remove all but k nodes).

Base Case: $k = 0, 1$ are meaningless, with no paths available. So remove all but 2 vertices from G . If they are connected, $m(G) = n(G) = 1$. If not, $m(G) = n(G) = 0$. The condition holds.

Inductive Case: Say for any induced subgraph $G^* = (V^*, E^*)$ of G , where $|V^*| = k - 1$, $n(G^*) \geq m(G^*)$.

If adding a node v (and its edges incident to V^*) back into G^* does not increase the number of vertex-disjoint paths, then our condition holds, since removing v (adding it to the cut set) doesn't change $n(G)$ or $m(G)$.

If doing so *does* increase the number of vertex-disjoint paths, then it can do so only by one: Cutting v yields G^* , where the condition holds, and adding v may increase the cut set by one, but can only add one more vertex-disjoint path to G^* (informally, you can't use v more than once). $\square$

Therefore $m(G) = n(G)$.

2 König's Theorem

Theorem 2.1 (König's Theorem). *In a bipartite graph, the size of the maximum matching equals the size of the minimum vertex cover (a set of nodes that together touch every edge).*

Proof. Assume we have nodes $\{a_i\}$ in one independent set and $\{b_i\}$ in the other independent set of the bipartite graph.

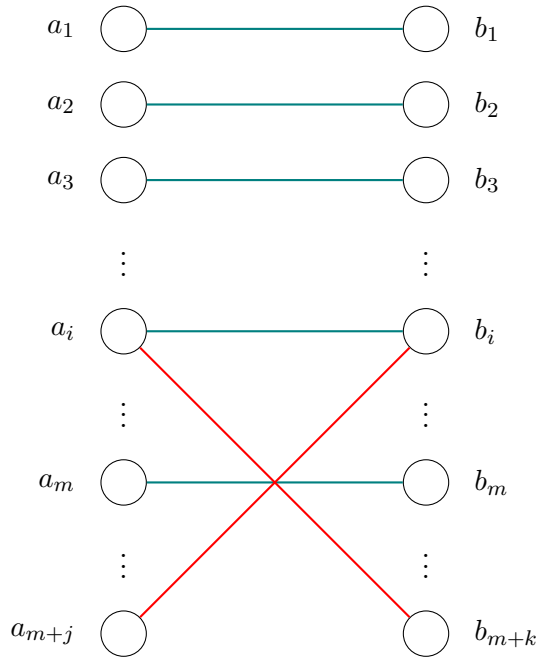
Relabel them so our maximum matching is $(a_1, b_1) \dots (a_m, b_m)$. This means there are sets of nodes $\{a_{m+i}\}$ ($i > 0$) that only connect to $b_1 \dots b_m$ and nodes $\{b_{m+i}\}$ ($i > 0$) that only connect to $a_1 \dots a_m$.

If some $a_{m+j}, j > 0$ connects to $b_i, 1 \leq i \leq m$, then no $b_{m+k}, k > 0$ can connect to a_i , since then we would have a bigger matching by removing (a_i, b_i) and adding (a_{m+j}, b_i) and (a_i, b_{m+k}) , a contradiction of our assumptions.

Therefore, for every node in the a -component outside the matching, add to the cover any adjacent b_i in the matching (and vice versa for every b node outside the matching). And there can be no nodes in the b -component outside the matching connecting to a_i , so only one of a_i, b_i will be selected. If a_i and b_i have no adjacencies outside the matching, or have not been selected after all the non-matched nodes are covered, then select either one for the cover. Therefore, we will have exactly one node in the cover for every edge in the matching.

This vertex cover covers all nodes outside the original matching and clearly covers all the nodes $a_1 \dots a_m, b_1 \dots b_m$ too. It is the minimum as any node set smaller than size m cannot cover the nodes in the matching.

□



3 Brooks's Theorem - TODO

Theorem 3.1. *If a graph G is not a clique and not an odd cycle and has maximum degree d , it can be properly colored with d colors.*

First, some lemmas. Call graphs which are not themselves odd cycles or complete graphs *allowed graphs*.

Note: TODO - if $d = 2$ at any point, it's a set of components, each of which are cycles. So these fit Brooks right away. Where can we assume $d \geq 3$?

Lemma 3.2 (Augmented odd cycle). *If an allowed graph G consists of an odd cycle C_n plus an additional node v , it is colorable with maximum degree $= d$ colors. Proof: $G - v = C_n$ is colorable in three colors, an alternating chain of $\frac{n-1}{2}$ red, blue pairs, plus a final green node. Since G is an allowed graph, v is connected to C_n , and therefore some node in C_n has degree ≥ 3 .*

If $\deg(v) \geq 4$, color v yellow, and $\chi(G) = 3 < d$. If $\deg(v) \leq 2$, a color is clearly available, so $\chi(G) = 3 = d$. If $\deg(v) = 3$, we consider first $|V| = 4$. This is disallowed since G would be a complete graph. Then since $|V| \geq 5 \rightarrow n \geq 4$, there is a node unconnected to v . Rotate the colors of C_n so this one is green and color v green. Then $\chi(G) = 3 = d$.

Lemma 3.3 (Augmented complete graph). *If an allowed graph G consists of an odd cycle K_n plus an additional node v , it is colorable with maximum degree $= d$ colors. Proof: K_n is clearly colorable in n colors. Since $G = (V, E)$ is an allowed graph, $0 < \deg(v) < n$, and v is connected to K_n in G . This means some node in K_n must have degree n . Additionally, since G is an allowed graph, there must be some node w , $(v, w) \notin E$ with degree $= n - 1$. Color v the same as w . The graph is colorable in n colors, with max degree n .*

Proof of Brooks's theorem:

Note: We assume this is a connected graph; if it's a disconnected graph, then the maximum degree of the component will be less than or equal to that of the graph, and we can color each component separately by the procedure below.

We will take any allowed G and proceed by induction on induced subgraphs $G^* = (V^*, E^*)$ of G .

Base Case: $|G^*| = 1$: The graph is a single node, a clique of size 1, colorable with one color.

Inductive Case: Assume for all induced subgraphs of G^* of $G = (V, E)$, the proposition holds and G is not an odd cycle or a complete graph and has max degree d .

Remove a node v of lowest degree in G . Consider the induced graph $G^* = G - v$, with max degree d , colorable in d colors.

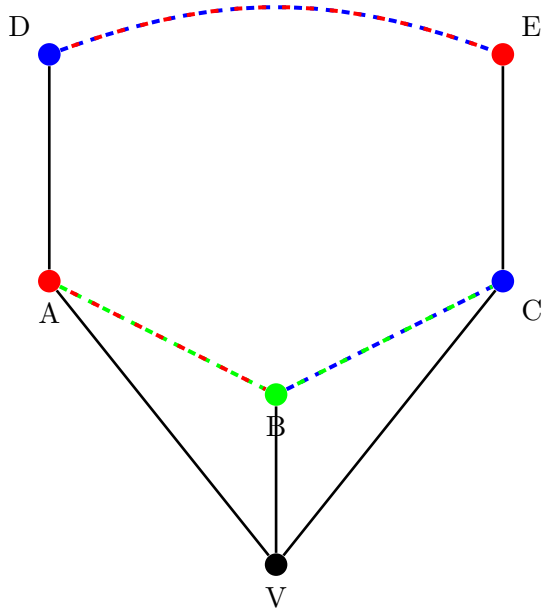
First, we assume $d \geq 3$, since otherwise we are done:

- If $d = 1$, we have a set of pairs. If $\deg(v) = 1$, G is 2-colorable. If $\deg(v) = 2$, and it is connected to two nodes in a pair, we have a disallowed subgraph. If $\deg(v) = 2$ and it is connected to two different pairs, that path is two-colorable. If $\deg(v) \geq 3$, color the pairs red-blue and v green.
- if $d = 2$, and does not include an odd cycle component (handled below), then including v brings d to 3. Color the even cycles red-blue and v green.
- If the induced graph is an odd cycle, then G is colorable in 3 colors by the Augmented Odd Cycle lemma.
- If the induced graph is a complete graph, then G is colorable by the Augmented Complete Graph lemma.
- If the maximum degree d of G^* is greater than $\deg(v)$, color the induced graph with d colors, and there will be at least one left for v .

We assume every node in G^* has degree at most v (TODO, can we have all degree 4, and removing a degree 4 makes it all degree 3?)

We will illustrate this with $d = 3$ in G^* (thus $d \leq 4$ in G) and then argue this generalizes to $d > 3$ in G^* .

Include v , uncolored, back in the graph as shown:



Our goal is to change the number of colors among $\{A, B, C\}$ to less than three. If that is the case, then v has a free one to select and we're done.

More assumptions:

- If two of v 's neighbors share a color already, then we're done. Therefore, assume v 's neighbors A , B , and C are colored red, green, and blue, respectively.
- (*) We show paths between A and C (ignoring D and E for the moment), A and B , and B and C , possibly just a single edge. They cannot all be single edges, however. If so, $G = \{A, B, C, V\}$ was a complete subgraph K_4 . If disconnected from the rest of the graph, we have a disallowed condition. If one of these nodes shares an edge outside of this set with some node w , then the max degree of G^* is 4 and v can be colored yellow. So at least one of the three paths has more than one edge (and since alternating colors, must have at least three edges).
- If there is not an alternating red-blue path from A to C , then switch red and blue along that path and all transitive neighbors. The graph will still be valid, with A and C both blue. Color V red.
- The same logic applies for red-green paths between A and B , and blue-green paths between B and C .
- So assume there are these three alternating colors paths as shown in the figure.
- (*) Note that paths between A and B , B and C , and A and C can be possibly just a single edge. They cannot all be single edges, however. If so, $G = \{A, B, C, V\}$ is a complete subgraph K_4 . If disconnected from the rest of the graph, we have a disallowed condition. If one of these nodes shares an edge outside of this set with some node w , then the max degree of G^* is 4 and v can be colored yellow. So some path needs to have at least four nodes instead of two. Without loss of generality, we'll pick the red-blue path, which must include nodes D and E beyond A and C .

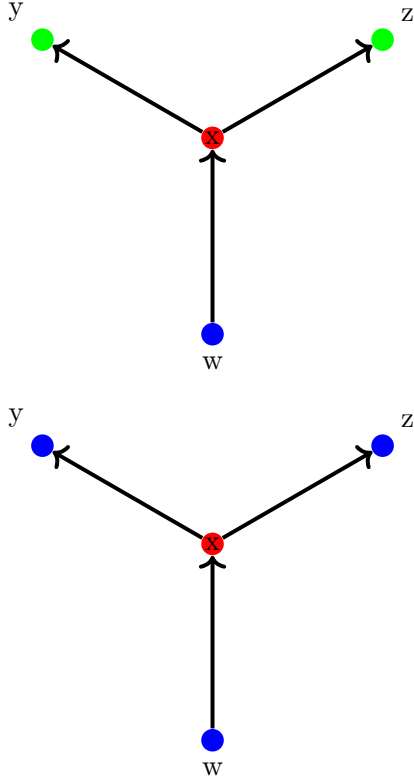
If any one of these three paths (A,B) , (B,C) , (A,C) can be flipped without flipping both endpoints, we can color v .

We introduce another other lemma to help us do our flipping:

Lemma 3.4 (Splitting Lemma). *If an alternating color path starting from some node v and proceeding $v_1, v_2, \dots$ encounters a node v_i where two successive nodes share a color, then the transitive chain starting from $v \dots v_i$ can be flipped.*

Proof: If in, say a red-blue chain, v_{i-1} is transformed to red, v_i has been transformed to blue, and $v_{i+1,1}$ and $v_{i+1,2}$ are both green, the chain must stop due to the degree of v_i . If $v_{i+1,1}$ and $v_{i+1,2}$ are both blue, change v_i to green. The whole sequence $v \dots v_i$ can

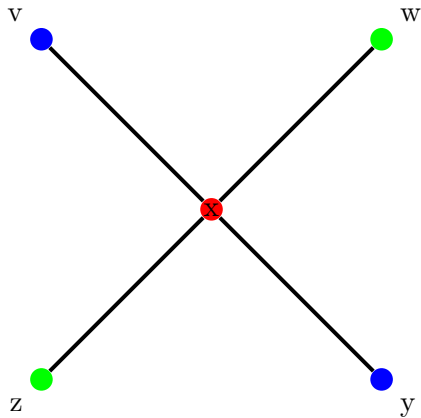
then be flipped (with v_i ending up in two possible valid states. In either case, v has been flipped.



In particular, this means that if chains of two colors SPLIT into child chains, the original chain can be flipped up to the point of intersection.

Corollary 3.5 (No crossing Lemma). *If two alternating chains meet, they cannot both continue from the point of intersection without one being flippable.*

Similarly to the splitting lemma, this means there are four node adjacent to the point of intersection, and therefore at least two colors unrepresented in the neighbor set, and the intersection can be colored with one of these.

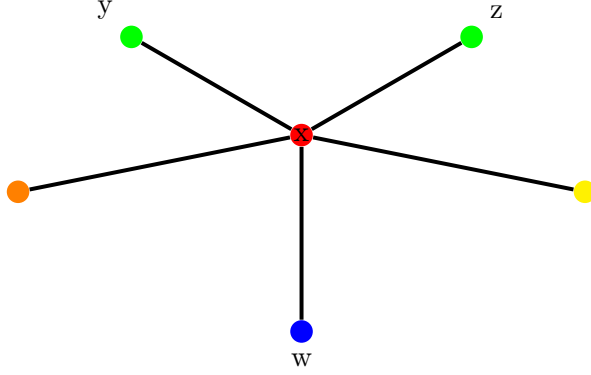


Coloring the figure

Consider the figure again, and consider a blue-green chain from node D.

- There must be a blue-green chain (of some positive length) from D, otherwise D has three red neighbors, and the splitting lemma applies when trying to draw a red-blue chain from A to C.
- That blue-green chain cannot intersect the chain from B to C, otherwise the splitting lemma applies when trying to draw a red-green path from B to C.
- If the blue-green chain does not intersect the red-green chain from A to B (or splits on the way, meaning D can be made green), A has two green neighbors and can be colored blue. Color v red.
- If the blue green chain DOES intersect the red-green chain from A to B at some formerly, green node w , and if w is not adjacent to A, A still has two green neighbors and can be made blue, and v red.
- If in the last case, w is adjacent to A, then A has a blue neighbor and a green neighbor. However, C now either connects to A (through E) through a split path, or not at all. So C can be flipped to red, and V to blue.

When flipping a path starting from A, B, or C, if the number of colors available is less than d , we can always end the chain there. This means, no matter how many colors beyond three we have, if a chain can "split", or if every node doesn't have $d - 1$ colors adjacent to it (excluding its own color), then we can stop our path flipping at that spot and convert the color of origin (in our case, A, B, or C).



Therefore, even if $d = 3 + j, j > 0$, we can ignore all of the j colors outside of these three when constructing these paths. The argument for at least one cycle of degree greater than three (marked (*) above) holds as well, since complete graphs of degree d follow the same argument

4 Lovasz's Theorem and Cobipartite Graphs

Definition 4.1 (Perfect Graph). *A weakly perfect graph G has a chromatic number $\chi(G)$ (the number of colors required for a vertex coloring) equal to its clique number $\omega(G)$ (the size of its largest fully-connected subgraph or clique). A refinement of this, a strongly perfect graph G , has this property for every induced subgraph $G^* \subset G$.*

Theorem 4.2 (Lovasz). *The complement of any perfect graph is perfect.*

Let's start with a seemingly smaller theorem: A cobipartite graph (complement of a bipartite graph) is perfect. Since a bipartite graph is clearly perfect (chromatic number of two, clique number of two), showing this proves Lovasz's theorem for a subset of graphs.

In a cobipartite graph, there are two cliques, K_m and K_n , with some set of connections between them.

Denote the maximum clique $C_m \cup C_n$, where $C_m \subseteq K_m, C_n \subseteq K_n$, and call M the remaining nodes in K_m . These are the nodes we need to color in $|C_n|$ colors to show $K_m \cup K_n$ is a weakly perfect graph.

Define the fanout $F(S)$ of a subset of $S \subseteq M$ as the set of nodes in M that are NOT connected to C_n . (This is an odd name, but it defines the set of colors M can take).

Lemma 4.3 (Expanding fanout lemma). *$|F(S)| \geq |S|$ for all $S \subseteq M$.*

Proof: Suppose that $|F(S)| < |S|$ for some $S \subseteq M$. This means that S is connected to all of C_m (since $M \subset K_m$), and everything in C_n except some set $F(S)$. However, the clique

$S \cup C_m \cap (C_n - F(S))$ is larger than $C_m \cap C_n$ since $|F(S)| < |S|$, which contradicts the maximality of $C_m \cap C_n$.

This also implies that $|F(M)| \geq |M|$, so $|C_n| \geq |M|$.

Lemma 4.4 (Fanout colorability lemma). *For every such $S \subseteq M$ with fanout $F(S)$ where $|F(S)| \geq |S|$, S can be colored with the colors of $F(S)$.*

We proceed by induction.

Base case: If $|S| = 1$, then there is at least one node in C_n unconnected to S , and we color it trivially.

Inductive step: The inductive step will be by contradiction. Suppose this is true for all $|S| < k$ but NOT true for $|S| = k$. First, if there is a subset of T of S where $|T| = |F(T)|$, then $|F(S - T)| \geq |S - T|$, and we have two smaller sets we can color separately by inductive hypothesis. Thus, we assume $|F(T)| > |T|$ for all $T \subset S$.

If T and $S - T$ both have fanouts larger than they, then they have to intersect in some node $X \in F(S)$. Color this node in T the color of the corresponding node in C_n (remember, they are the ones NOT connected to C_n) and remove it from T and $S - T$. These reduced sets are colorable by inductive hypothesis.

Therefore, M can be colored with the colors of C_n , and by symmetry, N can be colored with the colors of C_m .

Lovasz: Say a graph is perfect and colorable in k colors. This is actually a k -partite graph, where each partition corresponds to one of the k colors. Its complement has cliques $K_1 \dots K_k$ with some maximal clique among them.

The logic of the cobipartite graph applies completely; when discussing some M excluded from the maximal clique, it did not matter how C_n was distributed among other partitions. In a bipartite graph, they were all in a second partition, but in a k -partite graph, they could just as easily be spread among $k - 1$ partitions. So each excluded subset M_i of a partition can be colored with the colors of its fanout (among all $K_j, j \neq i$), no matter where they are grouped, and we have a perfect graph. We can say that a weakly perfect graph has a weakly perfect complement and by the Expanding Fanout Lemma, which will hold on every induced subgraph, a strongly perfect graph has a strongly perfect complement.

5 Turan's Theorem Original Proof

Define:

$$f(n, k) = \frac{1}{2} n^2 \left(1 - \frac{1}{k}\right).$$

Turan's theorem

Theorem 5.1 (Turan). *If a graph $G = (V, E)$ on n nodes satisfies $|E| > f(n, k)$, then G contains a $(k + 1)$ -clique.*

Contrapositive. If G has no $(k + 1)$ -clique, then

$$|E| \leq f(n, k).$$

Proof Plan: If we can transform any graph $G = (V, E)$ without a $(k + 1)$ -clique into a graph $G^* = (V, E^*)$ where G^* has no $(k + 1)$ -clique and $|E| = |E^*| \leq f(n, k)$, the contrapositive of Turan's theorem follows directly.

We consider such a set of graphs next: k -partite graphs. An example of these is bipartite graphs ($k = 2$).

Lemma 5.2 (k -partite bound). *A k -partite graph on n nodes contains no $(k + 1)$ -clique and satisfies*

$$|E| \leq f(n, k).$$

(The proof of the k -partite bound lemma is in a later section.)

Lemma 5.3 (Transformation lemma). *Let G be a graph on n nodes with no $(k + 1)$ -clique. Then there exists a k -partite graph on the same node set with the same number of edges, in particular:*

$$|E| \leq \frac{1}{2}n^2\left(1 - \frac{1}{k}\right),$$

Additionally, G possesses a node of degree at most $n(1 - \frac{1}{k})$.

Base case

For $n \leq k + 1$ the claim is immediate: the complete graph K_{k+1} contains $\frac{1}{2}(k + 1)k$ edges. If we have less than or equal to $\frac{1}{2}(k + 1)k - 1$ edges, we cannot form a $k + 1$ -clique. We show that this edge count is less than the bound.

$$-\frac{1}{2}k^2 - \frac{k}{2} + \frac{1}{2} \leq 0 \tag{1}$$

$$\frac{1}{2}k^3 + \frac{1}{2}k^2 - k \leq \frac{1}{2}k^3 + k^2 - \frac{1}{2}k - \frac{1}{2} \tag{2}$$

$$\frac{1}{2}k^3 + \frac{1}{2}k^2 - k \leq \frac{1}{2}(k^2 + 2k + 1)(k - 1) \tag{3}$$

$$\frac{1}{2}(k + 1)k - 1 \leq \frac{1}{2}(k + 1)^2 \frac{k - 1}{k} \tag{4}$$

(1) is clear since $k > 1$.

Then, any graph avoiding a $(k+1)$ -clique already satisfies the bound and is itself k -partite (some parts may be empty).

Because $|E| \leq \frac{1}{2}n^2(1 - \frac{1}{k})$, the average degree is at most $n(1 - \frac{1}{k})$; hence some node has degree at most $\lfloor n(1 - \frac{1}{k}) \rfloor$.

Inductive step

Assume $n > k+1$ and that the lemma holds for all graph of node count n or fewer. Let G be a graph on $n+1$ nodes with no $(k+1)$ -clique.

Every induced subgraph of size n also avoids $(k+1)$ -cliques and so meets the edge bound $f(n, k)$.

Goal: find a node v of degree

$$\deg(v) \leq D := n - \left\lfloor \frac{n}{k} \right\rfloor.$$

If so, we apply the induction hypothesis to $G - v$ to obtain a k -partite graph, then assign v to a smallest partition (necessarily of size $\leq \lfloor \frac{n}{k} \rfloor$), rewiring its at most D edges across the remaining partitions. The resulting graph is k -partite with the same edge count as G .

Lemma 5.4 (Degree bound lemma). *The above-mentioned bound $\lfloor n(1 - \frac{1}{k}) \rfloor$ is equal to the bound target bound D if $k \mid n$ and equal to $D - 1$ if $k \nmid n$.*

If $k \mid n$ then

$$n - \left\lfloor \frac{n}{k} \right\rfloor = \left\lfloor n \left(1 - \frac{1}{k} \right) \right\rfloor,$$

the floor operation becomes the identity and equality follows.

If $n = qk + r$, $\lfloor n(1 - \frac{1}{k}) \rfloor = \lfloor (qk + r) - q - \frac{r}{k} \rfloor$, and $n - \lfloor \frac{n}{k} \rfloor = qk + r - \lfloor (q + \frac{r}{k}) \rfloor$.

Taking out the integer $qk + r$, $\lfloor -q - \frac{r}{k} \rfloor < -\lfloor q + r/k \rfloor$, and they lie on either side of integer $-q$, so $\lfloor n(1 - \frac{1}{k}) \rfloor = n - \lfloor \frac{n}{k} \rfloor - 1$.

Existence of a small degree node

- If $k \nmid n$, choose any subset of n nodes of G , excluding some node v ,

By induction the subset contains a node w of degree $\leq n(1 - \frac{1}{k})$, which is $\leq D - 1$ by the Degree Bound Lemma, so the whole graph has such a node. Even if $(v, w) \in E$, w satisfies the degree bound.

- Suppose now $k \mid n$ (no more floors!) and, toward contradiction, that every node satisfies $\deg(v) \geq D + 1$.

Count edges in all $\binom{n+1}{n} = n + 1$ induced n -node subgraphs. By induction each has at most

$$f(n, k) = \frac{1}{2}n^2\left(1 - \frac{1}{k}\right)$$

edges. Summing and dividing by the multiplicity $n - 1$ with which each edge is counted (two subgraphs will contain its endpoints), we obtain an upper bound

$$T_{\text{upper}} = \frac{n+1}{n-1}f(n, k).$$

On the other hand, our degree assumption forces at least

$$T_{\text{lower}} = \frac{n+1}{2}\left(n - \frac{n}{k} + 1\right)$$

edges in total. Compute

$$T_{\text{lower}} - T_{\text{upper}} = \frac{n+1}{2} \frac{n-k}{k(n-1)} > 0$$

because $n > k$. This contradiction establishes that a node of degree $\leq D$ exists.

Completion of the inductive step

Removing the low-degree node and applying the rewiring argument yields a k -partite graph on $n + 1$ nodes with the same edge count as G . Hence every $(n + 1)$ -node graph without a $(k + 1)$ -clique satisfies

$$|E| \leq \frac{1}{2}n^2\left(1 - \frac{1}{k}\right).$$

Corollary (Turan). If $|E| > \frac{1}{2}n^2(1 - \frac{1}{k})$ then G must contain a $(k + 1)$ -clique.

5.1 Proof of k -partite lemma

A k -partite graph has k partitions of nodes, within which all are disconnected. The nodes from differing partitions may connect. Clearly, a $k + 1$ -clique cannot exist among them, as nodes within each partition are disconnected, and there are only k partitions.

A perfectly balanced graph has partitions all of equal size.

Proposition 5.5. *The edge-count formula of a perfectly balanced k -partite graph is*

$$E_p(n, k) = \frac{n^2}{2}\left(1 - \frac{1}{k}\right).$$

Each of n nodes connects to $n\frac{k-1}{k}$ nodes in other partitions. We divide by two to get the number of edges.

Proposition 5.6. *The edge-count formula of a perfectly balanced graph of node total n is the upper bound to any partitioning of nodes among k partitions.*

We are looking at maximizing $f(\vec{a}) = \frac{1}{2} \sum_{i \neq j; i, j < k}^k a_i a_j$, where a_i is the node count of partition i .

We can write this as

$$f(\vec{a}) = \frac{1}{2} \sum_{i \neq j; i, j < k}^k a_i a_j = \left(\sum_i^k a_i \right)^2 - \frac{1}{2} \sum_i^k a_i^2 = n^2 - \frac{1}{2} \sum_i^k a_i^2 \quad (5)$$

So we are looking at minimizing the final term to maximize $f(\vec{a})$

Consider the definition of statistical variance:

$$Var(a_1, a_2 \dots a_k) = \sum_{i=1}^k (a_i - \bar{a})^2 = \sum_{i=1}^k (a_i^2 - 2\bar{a}a_i + \bar{a}^2) \quad (6)$$

So $Var(a_1, a_2 \dots a_k) = \frac{1}{k} [\sum_{i=1}^k (a_i^2) - 2\bar{a}(\sum_{i=1}^k a_i) + k\bar{a}^2]$

Note that $2\bar{a}(\sum_{i=1}^k a_i) = 2k\bar{a}^2$.

Thus, ignoring constant terms, minimizing $Var(a_1, a_2 \dots a_k)$ is the same as minimizing $\sum_{i=1}^k a_i^2$. But the variance reaches its minimum at zero when all values a_i are equal. This is therefore bounded by the edge count of the equal partition $E_p(n, k) = \frac{n^2}{2} \left(1 - \frac{1}{k}\right)$.

5.2 Conclusion

Therefore, any k -partite graph has less than or equal to $\frac{n^2}{2} \left(1 - \frac{1}{k}\right)$ edges.